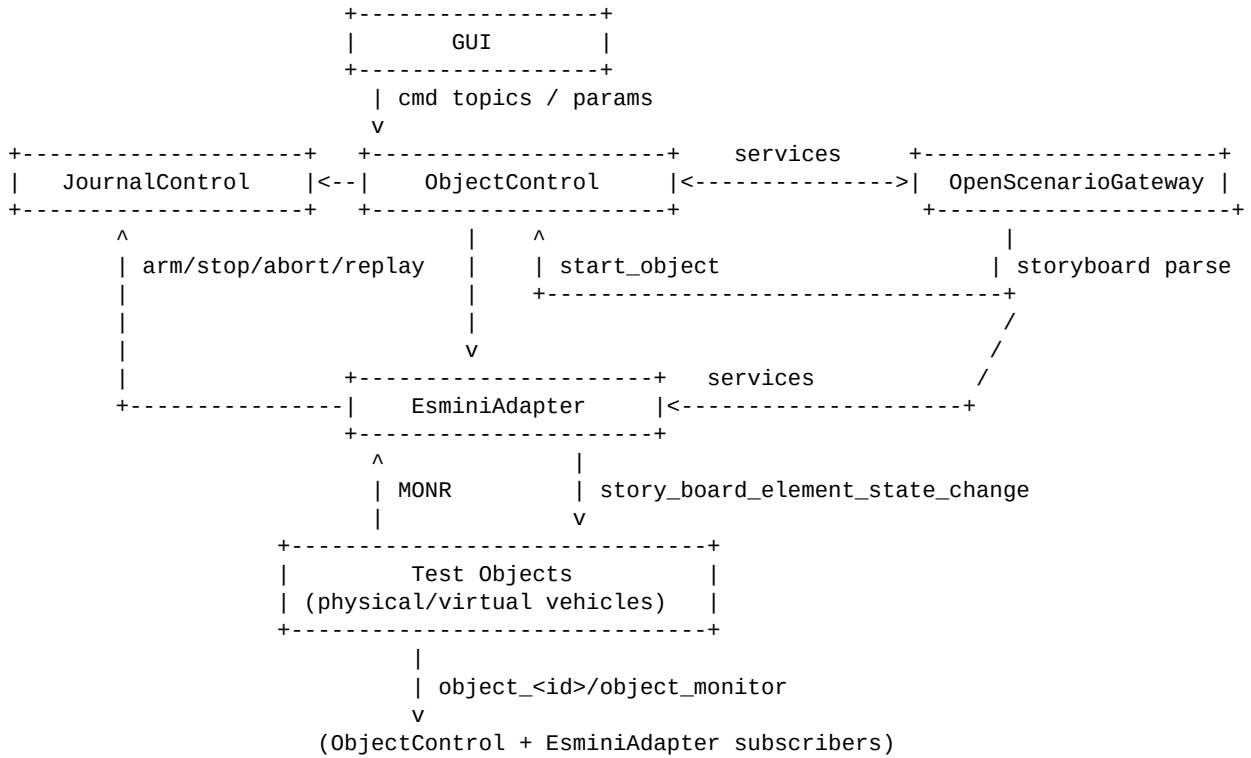


ATOS Module Interaction Diagrams

Generated from local ATOS code layout and launch profiles.

1) launch_basic.py Runtime Graph

Nodes: GUI, OpenScenarioGateway, ObjectControl, JournalControl, EsminiAdapter, and websocket bridge. ObjectControl is the orchestrator.



Bridge side path: GUI <--> FoxgloveBridge / Rosbridge

Core basic edges

- OpenScenarioGateway provides get_object_ids/get_object_ip/set_object_ip/get_open_scenario_file_path
- EsminiAdapter provides get_object_trajectory/get_test_origin
- ObjectControl consumes both service sets, publishes state_change, objects_connected, connected_object_ids

2) launch_full.py Runtime Graph (Basic + Add-ons)

launch_full.py extends basic with graphical + experimental nodes.

Base graph (GUI + OpenScenarioGateway + ObjectControl + JournalControl + EsminiAdapter + bridge) plus:

```
+-----+ +-----+
| PointcloudPublisher|<--init--|   ObjectControl   |
+-----+ +-----+
|
| +--> pointcloud topics
|
+-----+ get_object_ids +-----+
| TrajectoryletStreamer|----->| OpenScenarioGateway |
+-----+ get_object_trajectory
| +-----+
| +-----> EsminiAdapter
|
+-----+ <--- connected_object_ids --- ObjectControl
|   OSIAdapter   | <--- object_<id>/object_monitor --- vehicles
+-----+ ---- OSI stream ----> external consumer
|
+-----+ <--- custom_command_action --- OpenScenarioGateway
|   RESTBridge   | ---- HTTP POST -----> external endpoint
+-----+
|
+-----+ <--- custom_command_action --- OpenScenarioGateway
|   MQTTBridge   | <--- state_change ----- ObjectControl
+-----+ <--> MQTT broker/external systems
|
+-----+ <--- object_monitor (aggregate) --- object data
|   MonrRelay    | ---- UDP relay -----> remote host
+-----+
|
+-----+ <--- object_monitor (aggregate)
|   SqlBridge     | <--- get_open_scenario_file_path --- OpenScenarioGateway
+-----+ ---- PostgreSQL writes -----> DB
|
+-----+ <--- get_object_return_trajectory client --- ObjectControl
|   BackToStart   | ----> return trajectory service response ---->
+-----+
```

Not in launch_full by default

- SystemControl (legacy command server path) - DirectControl - IntegrationTesting (enabled by launch_integration_testing.py)

Generated file

/home/sepast/Documents/repos/simsvat/ATOS_module_interaction_diagrams.pdf